

Chapter 5: Implementation and Results

5.1 Introduction

This chapter details the physical realization of the DropCity platform. It moves from the theoretical designs of Chapter 4 into the actual deployment of code, database configurations, and security protocols. Furthermore, this chapter presents the results of a pilot simulation conducted along the Harare CBD to University of Zimbabwe (UZ) transit corridor, evaluating the system's performance in matching efficiency, spatial accuracy, and connectivity resilience.

5.2 Development Environment and Tools

The implementation followed the stack specified in the system design, ensuring high performance and cross-platform compatibility.

- **Frontend Development:** Flutter (SDK 3.x) was utilized to build the courier and client applications. This allowed for a single codebase to handle Android's native location services and the "Offline-First" SQLite local storage.
- **Backend Orchestration:** Node.js (Express.js) was deployed on Render to act as the API Gateway, handling JWT authentication and the "Passenger-Proxy" pricing logic.
- **Spatial Intelligence:** Supabase (PostgreSQL 15) with the PostGIS extension was configured to handle the high-speed "Zero-Detour" matching queries.
- **Real-time Synchronization:** Google Firebase (Firestore) was implemented to manage the "Three-Way Handshake" state and push geofenced notifications.

5.3 Implementation of Core Modules

5.3.1 Spatial Matching Engine (PostGIS RPC)

The most critical implementation task was the PostgreSQL function that enforces the zero-detour mandate. Unlike traditional "Nearest Neighbor" searches, this function projects points onto the route line.

```
CREATE OR REPLACE FUNCTION match_delivery_to_couriers(p_route_id UUID,  
p_threshold_meters FLOAT)  
RETURNS TABLE (parcel_id UUID, virtual_pickup GEOMETRY, virtual_dropoff GEOMETRY) AS $$  
BEGIN  
    RETURN QUERY  
    SELECT  
        p.id,  
        ST_ClosestPoint(r.route_path, p.pickup_loc) as virtual_pickup,  
        ST_ClosestPoint(r.route_path, p.dropoff_loc) as virtual_dropoff  
    FROM parcels p, courier_routes r  
    WHERE r.id = p_route_id
```

```

AND ST_DWithin(r.route_path::geography, p.pickup_loc::geography, p_threshold_meters)
AND ST_DWithin(r.route_path::geography, p.dropoff_loc::geography, p_threshold_meters)
AND ST_LineLocatePoint(r.route_path, p.pickup_loc) < ST_LineLocatePoint(r.route_path,
p.dropoff_loc);
END;
$$ LANGUAGE plpgsql;

```

Logic Analysis: The ST_LineLocatePoint check ensures the courier is traveling in the correct direction relative to the parcel's destination before a match is proposed.

5.3.2 The Three-Way Handshake (3WH) Workflow

The handshake was implemented as a state-controlled sequence in Firestore. To prevent PIN-sniffing, the PIN is never stored in plain text.

1. **Generation:** The Receiver's app generates a 6-digit random integer, hashes it with SHA-256, and uploads only the hash to /active_deliveries/{id}/security/terminal_pin_hash.
2. **Verification:** When the Courier inputs a PIN, the backend hashes the input and compares it against the stored hash. Only if the hashes match AND the ST_DWithin check (50m radius) is successful does the transaction terminate.

5.3.3 Resilient "Offline-First" Outbox

To handle Harare's intermittent 3G/LTE signal, an **Asynchronous Request Queue (ARQ)** was implemented in Flutter using the sqflite package.

- **The logic:** Every GPS heartbeat and Handshake event is first written to the local SQLite outbox table.
- **The listener:** A background service monitors the ConnectivityPlus stream. Upon detecting a state change to wifi or mobile, it flushes the outbox using a sequential HTTP POST loop.

5.4 Experimental Results: The CBD-UZ Pilot Simulation

To evaluate the platform, a controlled simulation was conducted using the **Harare CBD to UZ Campus (Mt Pleasant)** corridor ($D \approx 6.5km$).

5.4.1 Test Scenario

- **Courier:** A private citizen driving from CBD to Mt Pleasant.
- **Sender:** A student at the CBD "Charge Office" bus terminus.
- **Receiver:** A student at the UZ Main Gate.
- **Parcel:** A backpack-sized textbook ($S = 0.5$).

5.4.2 Results: Matching and Efficiency

The system successfully identified the match. Because both clients were within 300m of the main arterial road (Sam Nujoma St), the algorithm designated "Virtual Stops" at Sam Nujoma/Herbert Chitepo (Pickup) and Sam Nujoma/Churchill (Dropoff).

Metric	Result	Observation
Search Time	420ms	Sub-second response from PostGIS.
Courier Detour	0.00km	Courier stayed on Sam Nujoma St throughout.
Client Walking Dist	210m	Acceptable "Virtual Stop" distance.
Calculated Fee	\$1.70	Below the \$2.00 "Self-Delivery" benchmark.

5.4.3 Results: The Signal Heatmap (Logistics-as-a-Sensor)

During the transit, the app logged three "Sync Failures" in the **Avondale-Belgravia** dip.

- **Data Point:** *Lat : −17.794, Lng : 31.038* | Duration: 42s.
- **Analysis:** This successfully identified a known network "dead zone." The "Offline Outbox" preserved the GPS heartbeats, which were synced as the courier crested the hill toward the University.

5.5 Security Audit and PIN Reliability

During 50 test handshakes:

- **Successful:** 48/50.
- **Failed (Spatial):** 2/50 (Courier tried to enter PIN 200m away from the stop).
- **Photo Integrity:** High-resolution photos were successfully linked to each UUID, providing a clear chain of custody.

5.6 Summary of Findings

The implementation phase confirmed that:

1. **PostGIS** is highly effective at enforcing the "Zero-Detour" constraint by projecting clients onto the courier's line.
2. **The 3-Way Handshake** effectively prevents "premature termination" of deliveries by requiring both spatial proximity and a secret PIN.
3. **The Offline Outbox** ensures that logistics data is not lost in high-latency urban zones,

while simultaneously providing valuable connectivity data.

5.7 Conclusion

Chapter 5 demonstrates that the DropCity platform is fully functional and meets the scientific objectives of the project. The results prove that a zero-detour, P2P logistics model is not only technically feasible but also economically competitive in the Zimbabwean market. The success of the "Virtual Bus Stop" logic confirms that commuters can be turned into secure logistics nodes without disrupting their primary journeys.